

Akshat Agarwal

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Portfolio — GitHub — LinkedIn

EDUCATION

Vellore Institute of Technology, Vellore
B.Tech. in Information Technology

July 2024 – 2028
CGPA: 9.32/10

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, SQL

Computer Science: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Development: React.js, Next.js, Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, Redis, Supabase

Infrastructure and AI: Docker, Kubernetes, AWS, Git, Postman, RAG, LLMs, Browser Automation

EXPERIENCE

Sstudize Labs

Mar 2026 – Present

Software Development Lead; previously Software Developer Intern

Remote

- Lead software development and technical execution for an AI-enabled EdTech platform serving **500+ students**, coordinating feature development, code reviews, debugging, and releases.
- Designed and shipped full-stack features using **React, Next.js, Node.js, Express.js, PostgreSQL, and Supabase** across the software development lifecycle.
- Re-architected backend workflows and APIs for scalability and maintainability, reducing API response times by **35%** across assessment and test-generation workflows.
- Diagnosed production performance issues involving **connection pooling, request timeouts, authentication rate limits, and concurrent traffic** during **1000-VU stress testing**.

Samsung PRISM – Samsung R&D Institute India

May 2026 – Present

R&D Intern

Remote

- Developing an autonomous interaction framework using **LLM-driven browser agents** for dynamic web interfaces and multi-step workflows.
- Designed modular **agent orchestration** and adaptive interaction pipelines for dynamic DOM structures, interstitials, and anti-bot scenarios.
- Integrated browser automation, DOM analysis, LLM reasoning, and vision-model experimentation to improve reliability across dynamic site workflows.
- Evaluated **DOM-based and vision-based interaction strategies** for dynamic pages, improving agent adaptability across changing layouts and UI states.

Metis Intellisystems Pvt. Ltd.

May 2026 – Jun 2026

AI Developer Intern

Onsite

- Built an AI-assisted transaction intelligence pipeline processing **UPI, IMPS, NEFT, and RTGS** transactions across financial workflows.
- Developed a deterministic semantic classification engine covering **40+ transaction categories** using rule-based signals and transformer models.
- Improved classification consistency by **30%** through domain-specific feature extraction and hybrid classification workflows.
- Built a modular **transaction processing pipeline** for parsing, normalizing, extracting signals, and classifying heterogeneous financial transaction records.

PROJECTS

Bank Statement RAG – Financial Document Intelligence

GitHub

- Built a privacy-first **RAG pipeline** for querying bank statement PDFs using locally hosted **Qwen 2.5 7B** with Ollama.
- Designed custom document parsing, chunking, retrieval, and table-aware processing for accurate transaction and date-based queries.

ThinkBoard – Productivity Platform

GitHub

- Engineered a full-stack note management platform using **Next.js, Node.js, Express.js, MongoDB, and Redis**.
- Designed REST APIs and implemented **Redis-based rate limiting** to improve service stability under concurrent requests.

Campus Delivery Routing System

C, Graph Algorithms

- Implemented a campus delivery routing system using **graphs and Dijkstra's shortest-path algorithm** to compute efficient delivery routes.
- Implemented adjacency-list graph representation and optimized shortest-path computation using **Dijkstra's algorithm**, supporting weighted routes across campus locations.